

(iv) Existence of inverse:- $-a$ is inverse of a

$$a + (-a) = -a + a = 0$$

(v) Commutative:- let $a, b \in G$

$$a + b = b + a$$

$(\mathbb{Z}, +)$

~~$(\mathbb{Z}, +)$~~ is abelian group under addition.

Ques:- G is set of rationals except -1 binary operation $*$ is defined by $a * b = a + b + ab$. show that it is a group.

Sol:- ~~let a, b~~

3) closure:- let $a, b \in G$

$$a * b = a + b + ab \in G$$

closure is satisfied.

for Associative:-
(iv) let $a, b, c \in G$

$$a * (b * c) = (a * b) * c$$

$$\text{LHS} = a * (b * c)$$

$$= a * (b + c + bc)$$

$$= a + (b + c + bc) + a(b + c + bc)$$

$$= a + b + c + bc + ab + ac + abc$$

$$= a + b + c + ab + bc + ac + abc$$

$$\text{RHS} = (a * b) * c$$

$$= (a + b + ab) * c$$

$$= (a + b + ab) + c + (a + b + ab)c$$

$$= a + b + ab + c + ac + bc + abc$$

$$= a + b + c + ab + bc + ac + abc$$

$$\text{LHS} = \text{RHS}$$

Associativity is satisfied

(ii) identity:- $e * a = a * e = a$, e be identity element
 $e * a = a$

$$e * a \Rightarrow e + a + ea = a$$

$$e + ea = 0$$

$$e(1+a) = 0$$

$$\boxed{e = 0}$$

$$\boxed{a \neq -1}$$

So identity is satisfied.

(i) Inverse:- let $a \in G$, and a^{-1} is inverse of a .

$$a * a^{-1} = a^{-1} * a = e$$

$$a + a^{-1} + aa^{-1} = a^{-1} + a + a^{-1}a = 0$$

$$a + a^{-1} + aa^{-1} = a^{-1} + a + a^{-1}a$$

inverse is satisfied

$$a + a^{-1} + aa^{-1} = 0$$

$$a + a^{-1}(1+a) = 0$$

$$a^{-1}(1+a) = -a$$

$$\boxed{a^{-1} = \frac{-a}{1+a}}$$

$$a \neq -1$$

$(G, *)$ is a group.

Ques-3 In $\mathbb{Z}$ we define $a * b = a + b + 1$ show that $(\mathbb{Z}, *)$ is an abelian group.

Solⁿ i) Closure- let $a, b \in \mathbb{Z}$
 $a * b = a + b + 1 \in \mathbb{Z}$
closure is satisfied.

ii) Associative- let $a, b, c \in \mathbb{Z}$
 $(a * b) * c = a * (b * c)$

$$\begin{aligned} \text{LHS} &= (a * b) * c \\ &= (a + b + 1) * c \\ &= a + b + 1 + c + 1 = a + b + c + 2 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= a * (b * c) \\ &= a * (b + c + 1) \\ &= a + b + c + 1 + 1 \Rightarrow a + b + c + 2 \end{aligned}$$

$$\text{LHS} = \text{RHS} -$$

associative is satisfied.

iii) Identity element- let $a \in \mathbb{Z}$
 $e * a = a * e = a$

$$e + a + 1 = a + e + 1 = a$$

$$e + a + 1 = \cancel{a + e + 1} a \quad \because e = -1$$

$e + a + 1 = a + e + 1$
identity is satisfied

Qiv) Inverse:- let $a \in G$, a^{-1} is inverse of a .
 $a * a^{-1} = a^{-1} * a = e$

$$\begin{aligned} a + a^{-1} + 1 &= a^{-1} + a + 1 = e \\ a + a^{-1} + 1 &= a^{-1} + a + 1 = -1 \\ a + a^{-1} + 1 &= a^{-1} + a + 1 \\ \text{inverse is satisfied} \end{aligned}$$

$$\begin{aligned} a * a^{-1} &= a^{-1} * a = e \\ a * a^{-1} &= e \\ a + a^{-1} + 1 &= -1 \\ \boxed{a^{-1} = -2 - a} \\ -2 - a &\in \mathbb{Z} \end{aligned}$$

Qv) Commutative:-

let $a, b, e \in \mathbb{Z}$
 $a * b = b * a$

$$a + b + 1 = b + a + 1$$

$\therefore$ commutative is satisfied

$(\mathbb{Z}, *)$ is Abelian Group.

Qw- let G be the set of all positive rational numbers and $*$ be the binary operation on G defined by
 $a * b = \frac{ab}{7} \quad \forall a, b \in G$

Prove that $(G, *)$ is an abelian group.

Solve- $3 * x = 2 \rightarrow$ in G .

Qx- let $a, b \in G$

$\Rightarrow a * b = \frac{ab}{7} \in G$ Closure is satisfied

Qy- let $a, b, c \in G$

$$(a * b) * c = a * (b * c)$$

$$\text{LHS} = (a * b) * c = \frac{ab}{7} * c = \frac{abc}{49}$$

$$\text{RHS} = a * (b * c) = a * \frac{bc}{7} = \frac{abc}{49} \quad \text{Consistent}$$

Associative is satisfied

(ii) Identity element: let $a \in G$,

$$e * a = a * e = a$$

$$\frac{ea}{7} = \frac{ae}{7} = a$$

$$\frac{ea}{7} = \frac{ae}{7}$$

identity is satisfied

$$\frac{ae}{7} = a$$

$$\boxed{e = 7} \in G$$

(iv) Inverse: $a \in G$, a^{-1} is inverse of a .

$$a * a^{-1} = a^{-1} * a = e$$

$$\frac{aa^{-1}}{7} = \frac{a^{-1}a}{7} = 7$$

$$\frac{aa^{-1}}{7} = \frac{a^{-1}a}{7}$$

inverse is satisfied

$$\frac{aa^{-1}}{7} = 7$$

$$\boxed{a^{-1} = \frac{49}{a}}$$

(v) Commutative : let $a, b \in G$.
 $a * b = b * a$

$$\frac{ab}{7} = \frac{ba}{7}$$

$\therefore$ commutative is satisfied.

$(G, *)$ is abelian group.

$$\Rightarrow 3 * x = 21$$

$$\frac{3x}{7} = 21$$

$$\therefore a^{-1} = \frac{49}{a}$$

$$\frac{3x}{7} = \frac{49}{2}$$

$$3x = \frac{49 \times 7}{2}$$

$$x = \frac{49 \times 7}{6}$$

$$x = \frac{343}{6}$$

Ques - Show that the cube root of unity is an abelian group under multiplication

Note -

solⁿ

$$\omega = \sqrt[3]{1}$$

$$\omega = 1^{1/3}$$

$$\omega^3 = 1$$

$$\omega^3 - 1 = 0$$

$$(\omega - 1)(\omega^2 + \omega + 1) = 0$$

$$\text{or } \omega = 1,$$

$$\omega = \frac{-1 + i\sqrt{3}}{2}$$

$$\omega = 1, \frac{-1 + i\sqrt{3}}{2}, \frac{-1 - i\sqrt{3}}{2}$$

$$1, \omega, \omega^2$$

$\times$	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	$\omega^3 = 1$
ω^2	ω^2	$\omega^3 = 1$	ω

Cayley's table

$$\omega^4 = \omega^3 \cdot \omega = 1 \cdot \omega = \omega$$

$$G = 1, \frac{-1 + i\sqrt{3}}{2}, \frac{-1 - i\sqrt{3}}{2}$$

$$G = 1, \omega, \omega^2$$

is an abelian

Note:- $\omega\omega^2 = \omega^3 = 1$, $1 + \omega + \omega^2 = 0$

The above entries in the table satisfying all the axioms.

i) let $a, b \in G$,
 $a \cdot b \in G$
closure is satisfied.

ii) let $a, b, c \in G$
 $a \cdot (b \cdot c) = (a \cdot b) \cdot c$
 $abc = abc$
Associativity is satisfied.

iii) 1 is the top row indicate the identity
 $a \cdot 1 = 1 \cdot a = a$

Identity is satisfied.

iv) let $\omega^{-1} = 1$, $\omega^{-1} = \omega^2$, $(\omega^2)^{-1} = \omega$

Inverse is satisfied.

v) let $a, b \in G$, $a \cdot b = b \cdot a$

Commutativity is satisfied.

$(G, *)$ is abelian Group under multiplication.

Cayley's table

$\omega^{-1} = \omega^2$
 $\omega^{-2} = \omega$

Q. 9 If $G = \{f_1, f_2, f_3, f_4\}$ of four functions defined

by $f_1(x) = x$, $f_2(x) = -x$, $f_3(x) = \frac{1}{x}$
 $f_4(x) = -\frac{1}{x}$ $\forall x \in \mathbb{R} - \{0\}$. $\wedge$ is an abelian group.

Group.

$$\therefore f \circ g = f(g(x))$$

Sol

$\circ$	f_1	f_2	f_3	f_4	f_1
f_1	f_1	f_2	f_3	f_4	f_2
f_2	f_2	f_1	f_4	f_3	
f_3	f_3	f_4	f_1	f_2	
f_4	f_4	f_3	f_2	f_1	
f_4					

$$f_1 \circ f_1 = f_1(f_1(x)) = f_1(x) = f_1$$

$$f_1 \circ f_2 = f_1(f_2(x)) = f_1(-x) = -x = f_2(x)$$

$$f_1 \circ f_3 = f_1(f_3(x)) = f_1\left(\frac{1}{x}\right) = \frac{1}{\frac{1}{x}} = x = f_3(x)$$

$$f_1 \circ f_4 = f_1(f_4(x)) = f_1\left(-\frac{1}{x}\right) = -\frac{1}{-\frac{1}{x}} = x = f_4(x)$$

$$f_2 \circ f_1 = f_2(f_1(x)) = f_2(x) = -x = f_2$$

$$f_2 \circ f_2 = f_2(f_2(x)) = f_2(-x) = x$$

$$f_2 \circ f_3 = f_2(f_3(x)) = f_2\left(\frac{1}{x}\right) = -\frac{1}{\frac{1}{x}} = -x = f_4$$

$$f_0 \circ f_1 = f_2(f_1(x)) = f_2(-\frac{1}{x}) = \frac{1}{x} = f_3$$

$$f_3 \circ f_1 = f_3(f_1(x)) = f_3(-\frac{1}{x}) = \frac{1}{x} = f_3$$

$$f_3 \circ f_2 = f_3(f_2(x)) = f_3(-x) = -\frac{1}{x} = f_4$$

$$f_3 \circ f_3 = f_3(f_3(x)) = f_3(\frac{1}{x}) = x = f(1)$$

$$f_3 \circ f_4 = f_3(f_4(x)) = f_3(-\frac{1}{x}) = -x = f_2$$

$$f_4 \circ f_1 = f_4(f_1(x)) = f_4(-\frac{1}{x}) = \frac{1}{x} = f_3$$

$$f_4 \circ f_2 = f_4(f_2(x)) = f_4(-x) = \frac{1}{-x} = f_3$$

$$f_4 \circ f_3 = f_4(f_3(x)) = f_4(\frac{1}{x}) = -x = f_2$$

$$f_4 \circ f_4 = f_4(f_4(x)) = f_4(-\frac{1}{x}) = x = f_1$$

closure:- let $a, b \in G$.

$$\Rightarrow a \circ b \in G$$

closure satisfied

Associative:- let $a, b, c \in G$

$$a \circ (b \circ c) = (a \circ b) \circ c$$

Associative is satisfied

Identity:- f_1 is the top row indicate the identity.

Identity is satisfied.

Inverse:- $\text{let}(f)^T = f$

$$(f_2)^T = f_2$$

$$(f_3)^T = 13$$

$$(f^*)^T = f^*$$

Inverse, is Satisfies

Commutativity:- let $a, b \in G$, $a \circ b = b \circ a$

Commutative is satisfied

$(\mathbb{C}, +)$ is an abelian group under the composition

Ques- Show that the fourth roots of unity is an abelian group under multiplication.

$$\underline{\underline{80}} \quad x = 4\sqrt{1}$$

$$x = \pm \sqrt{4}$$

$$\underline{x^4 = 1}$$

$$(x^2 - 1)(x^2 + 1) = 0$$

~~$x^2=1, x^2=-1$~~

$$x = \pm 1 \quad x = \pm \sqrt{-1} = \pm i$$

$$C_n = 1, -1, i, -i$$

x	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	-1	1
$-i$	$-i$	i	1	-1

Addition / Multiplication Modulo:-

Addition Modulo:- $+_m$ $a + m b = r$ where, $0 \leq r < m$

Multiplication Modulo:- $\times_m$ $a \times m b = r$ where $0 \leq r < m$

eg $5 +_4 4 = 1$ $5 + 4 \Rightarrow 4 \overline{) 9} 2$

$5 +_3 11 = 1$

$5 +_{11} 11 = 16$

$3 \overline{) 16} 5$

$3 \times_6 16 = 0$

$3 \times_{16} 16 = 48$

$6 \overline{) 48} 8$

Ques - Prove that $Z_4 = \{0, 1, 2, 3\}$ is an abelian group w.r.t addition modulo 4.

Ans $Z_4 = \{0, 1, 2, 3\}$
Cayley's table.

$+_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

$0^{-1} = 0$
 $1^{-1} = 3$
 $2^{-1} = 2$
 $3^{-1} = 1$

i) Closure? All the entries in table are the elements of $\mathbb{Z}_4$.

Therefore the closure axiom is satisfied.

ii) Associative:-

$$\text{Consider } (1+2)+3 = 3+3 = 2$$

$$1+(2+3) = 1+1 = 2$$

Therefore Associativity is satisfied.

iii) Identity:- The row headed by the element 0 is identical to top row, 0 is identity element and $0 \in \mathbb{Z}_4$.

iv) Inverse:- From the table it is clear -

$$0^{-1} = 0$$

$$1^{-1} = 3$$

$$2^{-1} = 2$$

$$3^{-1} = 1$$

So, Every element have inverse in the set $\mathbb{Z}_4$.

v) Commutative:- From the table it is clear that about the principal diagonal, the table is symmetrical. Therefore commutative axiom is satisfied.

$(\mathbb{Z}_4, +_4)$ is an abelian group.

Properties of Group:-

i) Theorem:- Uniqueness of Identity (The identity element is unique)

Proof:- let $(G, *)$ be a group having two identities e and e' .

$$e \text{ is identity of } G \Rightarrow ee' = e' \quad \text{--- (1)}$$

$$e' \text{ is identity of } G \Rightarrow ee' = e \quad \text{--- (2)}$$

ee' is unique element of G .

so $e = e'$ Hence proved

ii) The inverse of a element in a group is unique.

Proof:- let $(G, *)$ be a group having two inverse b and c in the group.

$$a^{-1} = b \Rightarrow ba = e = ab$$

$$a^{-1} = c \Rightarrow ac = e = ca$$

$$ba = e \Rightarrow (ba)c = ec$$

$$b(ac) = ec$$

$$b(\phi) = e/c$$

$$\boxed{b = c}$$

iii) if G is a group for, $a, b \in G$

$$a) (a^{-1})^{-1} = a$$

$$b) (ab)^{-1} = b^{-1}a^{-1} \quad (\text{Reversal law})$$

Subgroups:

A non empty subset H of a group G is called Subgroup of G if

i) H is stable (closed) for the composition defined in G i.e

$$a \in H, b \in H \Rightarrow a, b \in H$$

and,

ii) H itself is a group for the composition induced by that of G .

Proper and improper ^(or Trivial) Subgroups:-

Every group G of order greater than 1 has atleast two subgroups. Which are:-

i) G (itself)

ii) $\{e\}$ is group the the Identity alone.

The above two ~~gr~~ subgroups are known as Improper or Trivial subgroups.

→ A subgroup other than these two subgroup G proper subgroup.

$$\boxed{N \subset Z \subset O \subset R \subset C}$$

Theorem:-

If H a subgroup of a G , then:-

- The identity of H is the same as that of G .
- The inverse of any element a of H is the same as the inverse of same regarded as a element of G .
- The order of any element a of H is the order of a in G .

Proof:- ^{Q1} let e and e' are identity element of G and H respectively.

$a \in H$ then
 $ae' = e'a = a$ — ①

$a \in H \Rightarrow a \in G$
 $ae = ea = a$ — ②

By ① and ②

$$\begin{aligned} ae' &= ae \\ \boxed{e &= e'} \end{aligned}$$

b) b and c are inverse element of G and H
 $a \in H$ then

~~a~~ $ac = ca = e$ — ①

$a \in H \Rightarrow a \in G$
 $ab = ba = e$ — ②

By ① and ②
 $\boxed{b = c}$

c) let the order of $a \in H$ be m and $n \in H$ and C_n respectively, if e be the identity, then by the definition of order.

$$a^m = e \text{ and } a^n = e \Rightarrow a^m = a^n$$

$$a^m a^{-n} = a^n a^{-n} = a^0$$

$$a^{m-n} = a^0$$

$$m-n \geq 0$$

$$\boxed{m \geq n}$$

Therefore, the order of any element of sub group is the same as that of the sub group and the original group.

Theorem:- The intersection of any two subgroups of a group G is again a subgroup of G .

Proof:- let H_1 and H_2 be two subgroups of a group G .

$$e \in H_1, e \in H_2 \Rightarrow e \in H_1 \cap H_2 \Rightarrow H_1 \cap H_2 \neq \emptyset$$

Now, let $a, b \in H_1 \cap H_2$ then

$$a, b \in H_1 \cap H_2 \Rightarrow a, b \in H_1 \text{ and } a, b \in H_2$$

$$ab^{-1} \in H_1 \text{ and } ab^{-1} \in H_2 \quad [H_1 \text{ and } H_2 \text{ are subgroups}]$$

$$ab^{-1} \in H_1 \cap H_2$$

$H_1 \cap H_2$ is a subgroup of G .

Ques:- Show that the group addition

Cayley's

	+	6	
		0	
		2	
		4	

Ques:- Prove that of the

clearly

let $x, y \in$

$$x - y =$$

in H and
then by

Ques:- Show that $H = \{0, 2, 4\}$ is a subgroup of the group that $G = \{0, 1, 2, 3, 4, 5\}$ under addition modulo 6.

$G = \{0, 1, 2, 3, 4, 5\}$

$H = \{0, 2, 4\}$

$H \subset G$ — ①

Cayley's Table or Composition Table -

$+_6$	0	2	4
0	0	2	4
2	2	4	0
4	4	0	2

closure.

$0^{-1} = 0$

$2^{-1} = 4$

$4^{-1} = 2$

Ques:- Prove that $H = \{a + ib \mid a, b \in \mathbb{Q}\}$ is a subgroup of the group $(\mathbb{C}, +)$.

clearly H is non empty subset of $\mathbb{C}$

let $x, y \in H$ where $x = a_1 + i b_1$
 $y = a_2 + i b_2$

$x - y = (a_1 + i b_1) - (a_2 + i b_2)$
 $= (a_1 - a_2) + i(b_1 - b_2) \in H$

$\therefore [(a_1 - a_2) \in \mathbb{Q}, b_1 - b_2 \in \mathbb{Q}]$

$$x \in H, y \in H \Rightarrow (x-y) \in H$$

Therefore H is a subgroup of $(G, +)$

Cyclic Group:-

A Group G is a cyclic group if there exists an element $a \in G$ such that

$$G = [a]$$

i.e. Every element of G can be expressed as some integral power of a .

a is called the generator of G .

$$G = \{ \dots, a^{-3}, a^{-2}, a^{-1}, a^0, a^1, a^2, a^3, \dots \}$$

↑
for multiplication operation

If the operation of the group G is addition $(+)$ then

$$G = [a] = \{ \dots, -3a, -2a, -a, 0, a, 2a, 3a, \dots \}$$

Ques- Prove that the set $G = \{x \mid x^n = 1\}$ of n^{th} roots of unity is a finite multiplicative cyclic group of order n .

Sol-

Q.3) let $x_1, x_2 \in G$ then $x_1^n = 1$ and $x_2^n = 1$

But $x_1^n = 1, x_2^n = 1 \implies x_1^n x_2^n = 1 \cdot 1 = 1$

$(x_1 x_2)^n = 1$

$x_1 x_2 \in G$

$\therefore x_1 \in G, x_2 \in G \implies x_1 x_2 \in G$

i.e. G is closed for multiplication

Also, $1^n = 1 \implies 1 \in G$ which is the identity for multiplication

further for each $x \in G$.

$x \in G \implies x^n = 1 \implies \frac{1}{x^n} = 1 \implies \left(\frac{1}{x}\right)^n = 1 \implies \frac{1}{x} \in G$

which is the inverse of x . So each element of G is invertible.

lastly, the multiplication of numbers is associative so, it also associative in G . Hence G is group for multiplication.

$x^n = 1$ has exactly n roots. So $O(G) = n$.

G is cyclic Group:-

$$\therefore 1 = \cos 2m\pi + i \sin 2m\pi \quad m \in \mathbb{Z}$$

$$1^{1/n} = \cos \frac{2m\pi}{n} + i \sin \frac{2m\pi}{n} \quad m = 0, 1, 2, 3, \dots, (n-1)$$

(By De Moivre's Theorem)

$$= e^{2m\pi i/n}$$

$m = 0, 1, 2, 3, \dots, (n-1)$ (By Euler's Theorem)

$$\therefore G = \{ e^{2\pi i/n}, e^{4\pi i/n}, e^{6\pi i/n}, \dots, e^{2m\pi i/n} \}$$

Therefore G is a cyclic group of order n

generated by $e^{2\pi i/n}$

Ques- Find all the generators of the cyclic group.
($G = \{0, 1, 2, 3, 4, 5, +_6\}$)

Sol-

$$1(0) = 0$$

$$\Rightarrow 0(0) = 1$$

$$1(1) = 1, 2(1) = 2, 3(1) = 3, 4(1) = 4, 5(1) = 5$$

$$6(1) = 0$$

$$\Rightarrow 0(1) = 6$$

$$1(2) = 2, 2(2) = 4, 3(2) = 0$$

$$0(2) = 3$$

$$1(3) = 3, 2(3) = 0$$

$$0(3) = 2$$

$$1(4) = 4, 2(4) = 2, 3(4) = 0$$

$$0(4) = 3$$

$$1(5) = 5, 2(5) = 1, 3(5) = 4, 4(5) = 2$$

$$5(5) = 3, 6(5) = 0$$

$$0(5) = 6$$

order of ~~element~~ ^{G} = 6.

Therefore $G = \langle 1 \rangle, = \langle 5 \rangle$

ie 1 and 5 are two generators of G .

Properties of cyclic Group:-

(i) Every cyclic Group is abelian.

Proof:- Let $G = \langle a \rangle$ be a cyclic group and $x, y \in G$,
Where,

$$x = a^m, y = a^n, m, n \in \mathbb{Z}$$

$$\begin{aligned} \text{Then } xy &= a^m a^n = a^{m+n} = a^{n+m} \\ &= a^n a^m = yx \end{aligned}$$

G is abelian Group.

Note Every cyclic group is abelian, but
Every abelian may not be cyclic.

(ii) If a is a generator of a cyclic group G ,
then a^k is also its generator.

Ques The order of a finite group is equal to the
order of its generator.

$$\text{i.e. } O(\text{finite cyclic group}) = O(\text{generator of group})$$

(iv) Every infinite cyclic group, has two
and only two generator.

(v) Every subgroup of cyclic group is cyclic.

Cosets:-

Let H be a subgroup of group G and $a \in G$, then the set $aH = \{ah \mid h \in H\}$ is called left coset of H in G .
and $Ha = \{ha \mid h \in H\}$ is called right coset of H in G .

By the Definition, it is clear that corresponding to every element of G , we have a left coset and right coset of H in G . It is obvious that

$$aH \subseteq G, Ha \subseteq G$$

Further we may note that, $eH = H = He$

i.e. the left and the right cosets of H corresponding to the identity element e coincide with H .

Hence H itself is a left as well as right coset of H in G .

Ques:- let $G = (\mathbb{Z}, +)$ and $H = 2\mathbb{Z} = \{0, \pm 2, \pm 4, \dots\}$
then the right coset of H in G are:

$$H+0 = \{0, \pm 2, \pm 4, \dots\} = H = 0+H = H+0$$

$$H+1 = \{\pm 1, \pm 3, \pm 5, \dots\}$$

$$H+2 = \{0, \pm 2, \pm 4, \dots\} = H$$

$$\begin{aligned} 2+2 &= 4 \\ -2+2 &= 0 \end{aligned}$$

It can be easily observed that any right coset and its corresponding left coset are equal i.e.

$$H+1 = 1+H, \quad H+2 = 2+H \quad \text{etc}$$

' $(\mathbb{Z}, +)$ is abelian group'.

$$H+3 = \{ \dots -6, -3, 0, 3, 6, 9, 12 \dots \} = H$$

$$H+4 = \{ \dots -5, -2, 1, 4, 7, 10, 13 \dots \} = H+1$$

Similarly, $H+5 = H+2$, $H+6 = H$

Thus H has only two distinct cosets $H, H+1$

Clearly, $G = H \cup (H+1)$

ex-2:- Find all the cosets of $3\mathbb{Z}$ in group $(\mathbb{Z}, +)$

Ans:- Let $H = 3\mathbb{Z} = \{ \dots -6, -3, 0, 3, 6, 9 \dots \}$

$$0+H = H+0 = \{ \dots -6, -3, 0, 3, 6 \dots \}$$

$$1+H = H+1 = \{ \dots -5, -2, 1, 4, 7 \dots \}$$

$$2+H = H+2 = \{ \dots -4, -1, 2, 5, 8 \dots \}$$

$$3+H = H+3 = \{ \dots -3, 0, 3, 6, 9 \dots \} = H+0$$

$$4+H = H+4 = \{ \dots -2, 1, 4, 7, 10 \dots \} = H+1$$

$$5+H = H+5 = \{ \dots -1, 2, 5, 8, 11 \dots \} = H+2$$

$$6+H = H+6 = \{ \dots 0, 3, 6, 9, 12 \dots \} = H+0$$